

LITERATURE REVIEW

Quantum Machine Learning for Early Disease Prediction

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1. Introduction

Quantum Machine Learning (QML) combines quantum computing concepts with machine learning methods and has become an active research area in healthcare. Existing studies have investigated QML for medical imaging, clinical decision support, disease classification, cardiovascular prediction, and digital health applications. The literature includes quantum support vector methods, quantum neural networks, variational quantum circuits, quantum kernels, and hybrid quantum–classical approaches.

The studies reviewed in this document were selected from the references used in the associated conference paper. The review focuses on the methods used, healthcare applications, reported findings, limitations, and directions for future research. Particular attention is given to cardiovascular and heart disease prediction because the proposed project implements a quantum-kernel-based heart disease classification workflow using the UCI Heart Disease dataset.

2. Review of Existing Studies

Study 1 — Ullah and García Zapirain (2024) [1]

The study presents a systematic review of the emerging applications of Quantum Machine Learning in healthcare. It discusses the use of QML across areas such as medical data analysis, disease-related applications, and healthcare decision support. The review highlights the potential of combining quantum computing with machine learning while also discussing practical limitations associated with current quantum hardware. Noise, limited qubit availability, and the maturity of quantum technology remain important barriers to large-scale healthcare applications. The study supports the need for larger datasets, stronger validation, and experiments on more realistic quantum platforms.

Study 2 — Alotaibi et al. (2023) [2]

This study investigates quantum-enhanced machine learning algorithms for heart disease prediction. It combines QML with preprocessing and quantum-based optimization and reports improved performance over the evaluated classical baseline in its experimental setting. The study demonstrates that heart disease is a relevant application area for quantum-enhanced classification. However, the reported performance depends on the selected dataset, preprocessing procedure, and computational environment. Future work should therefore test the approach on broader datasets and more realistic quantum computing environments.

Study 3 — Chow (2025) [3]

This review examines quantum computing and machine learning for medical decision-making. It discusses how quantum and classical learning techniques can support diagnosis, prediction, optimization, and clinical decision-support tasks. The study provides a broad view of the relationship between quantum computing and healthcare applications rather than focusing on a single disease dataset. It also indicates that many healthcare

applications remain experimental or conceptual, creating a need for real-world case studies and clinically validated implementations.

Study 4 — Upama et al. (2023) [4]

This survey reviews QML applications for disease detection and prediction across healthcare domains. It covers different QML approaches and emphasizes the growing interest in quantum methods for healthcare classification problems. The survey identifies practical concerns related to hardware maturity, data representation, and validation. It suggests that future research should develop reliable benchmarks, improve feature encoding, and evaluate QML models on quantum hardware rather than relying only on idealized simulation.

Study 5 — Prabhu et al. (2023) [5]

The QuCardio study investigates Quantum Machine Learning for cardiovascular disease detection. The work considers quantum support-vector approaches, Pegasos QSVC, and quantum neural architectures. A QNN-based model reported 97.31% accuracy for the studied multi-class ECG-image task, while the study also compared quantum classifiers with classical configurations. The results indicate that quantum methods can be explored for cardiovascular applications, but their performance must be interpreted in relation to the dataset and experimental configuration. The study motivates further investigation of quantum-kernel methods on structured clinical features.

Study 6 — Babu, Ramya, and Gracewell (2024) [6]

This study evaluates Quantum-Enhanced Machine Learning (QuEML) for heart disease prediction using a Kaggle heart disease dataset. The authors compare quantum-enhanced approaches with traditional machine learning using measures such as accuracy, precision, recall, specificity, F1-score, and training time. The reported results show a modest accuracy improvement and lower average training time for the proposed quantum approach in the study's experimental setting. Because the evaluation uses a particular dataset and implementation environment, the findings require validation on larger and more clinically representative datasets.

Study 7 — Cerezo et al. (2022) [7]

This work provides a broad discussion of the challenges and opportunities in Quantum Machine Learning. It addresses issues such as trainability, noise, scalability, data encoding, and the difficulty of demonstrating practical quantum advantage. Although it is not limited to healthcare, the study is important for interpreting healthcare QML results because many medical experiments use current noisy or simulated quantum environments. The work highlights the need for improved algorithms, error mitigation, scalable architectures, and reliable evidence of quantum advantage.

Study 8 — Gupta et al. (2025) [8]

This systematic review evaluates QML applications in digital health and compares quantum approaches with existing classical methods. The review found that only a small portion of the included studies considered realistic operating conditions involving quantum hardware or noisy simulations. It also reported that performance differences between quantum and classical models did not show a consistent trend supporting empirical quantum utility in digital health. The findings emphasize the importance of rigorous benchmarking, realistic hardware evaluation, scalability, and clinical validation.

3. Thematic Analysis of the Literature

- **QML approaches used:** The reviewed studies use several QML approaches, including QSVM/QSVC, Pegasos QSVC, QNN, VQC, quantum ensemble methods, and quantum kernels. Hybrid classical–

quantum workflows are common because classical methods can perform preprocessing and dimensionality reduction before quantum processing.

- **Healthcare and disease applications:** Applications cover general healthcare analytics, medical decision support, disease detection, cardiovascular disease, and heart disease prediction. Cardiovascular applications are particularly relevant to the present project because several reviewed studies directly investigate heart or cardiovascular disease.
- **Datasets and data representation:** The literature uses different forms of healthcare data, including structured clinical attributes, ECG-related data, and other digital-health datasets. The choice of representation affects the quantum model because classical healthcare features generally need preprocessing, scaling, and dimensionality reduction before being encoded into quantum circuits.
- **Evaluation and comparison:** Accuracy, precision, recall, F1-score, specificity, and training time are frequently used for evaluation. Comparison with classical baselines is important, but the literature does not always use identical training budgets, datasets, or experimental conditions.
- **Common limitations:** The most repeated limitations are small or specialized datasets, dependence on simulation, limited quantum hardware, noise, data encoding requirements, scalability, inconsistent benchmarking, and limited clinical validation.

4. Comparative Analysis

Table 1 summarizes the major characteristics of the eight reviewed studies. The comparison is intended to show the evolution of QML research in healthcare and the recurring gaps that motivate the present experimental work.

Ref.	Application / Focus	QML Method	Main Finding	Major Limitation
[1]	Healthcare / QML review	Multiple QML approaches	QML shows growing potential across healthcare applications.	Limited clinical validation and hardware maturity.
[2]	Heart disease	QML + QPSO	Reported improved performance over the evaluated classical model.	Dataset and computational setting limit generalization.
[3]	Medical decision support	QC + ML / QML	Reviews quantum methods for diagnosis, prediction, optimization, and decision support.	Many applications remain experimental or conceptual.
[4]	Disease detection and prediction	Various QML methods	Surveys QML applications across healthcare domains.	Hardware and practical validation remain limited.
[5]	Cardiovascular disease	QSVC, Pegasos QSVC, QNN	QNN reported 97.31% accuracy for the studied ECG-image task.	Results depend on dataset and experimental setup.
[6]	Heart disease	Quantum Ensemble Learning	Reported a modest accuracy improvement and reduced training time in its experiment.	Dataset and evaluation setting require broader validation.
[7]	QML theory and applications	QNN, QBoost, related methods	Identifies trainability, noise, scalability, and quantum-advantage challenges.	Hardware and scalability remain unresolved.
[8]	Digital health	Various QML models	Finds no consistent empirical evidence of quantum utility in realistic digital-health settings.	Few realistic hardware evaluations and inconsistent benchmarking.

Table 1. Comparative analysis of selected QML studies in healthcare and disease prediction.

5. Research Gap Identified

The literature shows that QML has been investigated for disease prediction and healthcare decision support, but the evidence is not yet sufficient to establish a general practical quantum advantage. Many studies are based on small or specialized datasets and rely on simulators rather than physical quantum processors. Differences in preprocessing, feature representation, model configuration, and classical baseline selection also make direct comparisons difficult.

Another important gap is the connection between a QML algorithm and a complete data-processing workflow. A practical experiment needs to include data cleaning, scaling, feature reduction, quantum representation, classification, and transparent evaluation. The reviewed cardiovascular studies provide motivation for using QML for heart disease, but there remains scope for controlled experiments that explicitly integrate classical preprocessing with a quantum-kernel classifier.

The present project addresses this gap by implementing a hybrid classical–quantum heart disease classification workflow using the UCI Heart Disease dataset. The approach reduces the 13 clinical features to four PCA components, applies a four-qubit ZZFeatureMap and Fidelity Quantum Kernel, and uses QSVC for classification. A classical SVC is evaluated using the same matched 120-sample training subset and the same 61-sample test set, providing a controlled comparison under the computational constraints of quantum-kernel simulation.

6. Relevance to the Present Project

The literature directly supports the technical direction of the project. Studies [2], [5], and [6] demonstrate the relevance of quantum-enhanced learning for heart or cardiovascular disease. Studies [1], [3], [4], [7], and [8] provide broader evidence about QML in healthcare and the challenges that must be considered when interpreting experimental results.

Based on these findings, the present project does not assume that a quantum model must outperform a classical model. Instead, it investigates whether a quantum-kernel workflow can be implemented reproducibly on structured clinical data and evaluates its behavior against a classical SVC under a controlled experimental setup. This positioning keeps the work focused on practical QML experimentation rather than claiming clinical deployment or demonstrated quantum advantage.

7. Literature Review Summary

The reviewed literature indicates that QML is an active and developing research area in healthcare. Quantum kernels, QSVC, QNN, VQC, and hybrid quantum–classical models have been explored for disease and cardiovascular prediction. The studies demonstrate promising experimental possibilities, but recurring limitations include dataset size, simulation dependence, hardware constraints, data encoding, scalability, benchmarking, and clinical validation.

These observations justify the development of a controlled experimental framework for heart disease classification. The present project contributes an implementation-oriented study that connects classical preprocessing and PCA with a quantum feature map, Fidelity Quantum Kernel, and QSVC, while retaining a matched classical SVC comparison.

8. References

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